

# Rendering Guide: QMD to HTML/PDF

This guide covers how to render all the .qmd files to HTML and PDF for deployment.

## Quick Start

```
cd website
quarto render
```

That's it! Quarto will: - Render all .qmd files in **content/** to HTML and PDF - Output to **../docs/** folder - Create standalone HTML files (with embedded CSS/resources) - Generate PDF versions - Generate PowerPoint presentation - Copy resources (images) as needed

## Before You Render

### Prerequisites

1. **Install Quarto** (if not already installed)

```
# macOS
brew install quarto

# Or download from: https://quarto.org/docs/get-started/
```

2. **Verify installation**

```
quarto --version
```

## What Gets Rendered

### .qmd Files → HTML + PDF

**Resources (pre-reading materials):** - website/content/what-is-ai.qmd → docs/what-is-ai.html + docs/what-is-ai.pdf - website/content/what-are-llms.qmd → docs/what-are-llms.html + docs/what-are-llms.pdf - website/content/quick-start-guide.qmd → docs/quick-start-guide.html + docs/quick-start-guide.pdf - website/content/what-to-expect.qmd → docs/what-to-expect.html + docs/what-to-expect.pdf - website/content/ai-in-pedagogical-design-and-delivery.qmd → docs/ai-in-pedagogical-design-and-delivery.html + .pdf - website/content/cognitive-prompting-in-education.qmd → docs/cognitive-prompting-in-education.html + .pdf

- website/content/using-ai-in-education-roles-for-teaching-and-research.qmd → docs/using-ai-in-education-roles-for-teaching-and-research.html + .pdf

**Materials (handouts, presentation, references):** - website/content/materials/teaching-strategies-handout.qmd → HTML + PDF - website/content/materials/references.qmd → HTML + PDF - website/content/materials/presentation.qmd → HTML + PDF + PPTX

### Assessment (already HTML)

- website/content/assessment/ai-literacy-diagnostic.html → (copied to docs/)

### Static Website (copied as-is)

- website/src/index.html → docs/index.html
- website/src/resources.html → docs/resources.html
- website/src/materials.html → docs/materials.html
- website/src/css/style.css → docs/css/style.css
- website/src/images/ → docs/images/

## Render Output Features

### HTML Files

- **Standalone:** All CSS and resources embedded (one file = complete)
- **Table of Contents:** Auto-generated from headings
- **Responsive:** Works on mobile and desktop
- **Searchable:** Can search within the page
- **Printable:** Print-friendly styling

### PDF Files

- **Professional:** Formatted for printing
- **Navigation:** Table of contents with clickable links
- **Color-coded:** Headings and links in colour
- **Metadata:** Title, author, date in document properties

## PowerPoint (PPTX)

- **Presentation-ready:** Use for live presenting
- **Slide per section:** Each top-level heading = one slide
- **Images embedded:** All presentation images included
- **Editable:** Customize in PowerPoint/Google Slides as needed

## Quarto Configuration

The rendering is configured in `website/_quarto.yml`:

```
project:
  type: default
  output-dir: ../docs      # Output to docs/
  resources:
    - src/images/          # Include images

render:
  - content/**/*.qmd      # Render all .qmd files

format:
  html:
    embed-resources: true  # Standalone HTML files
    toc: true             # Table of contents
    embed-resources: true
    css: src/css/style.css # Use website CSS

  pdf:
    toc: true             # PDF also has TOC
    colorlinks: true      # Coloured links in PDF

execute:
  freeze: auto            # Cache execution results
```

## Before Rendering

Check that: 1. All `.qmd` files have YAML front matter (between `---` lines) 2. Image paths in presentation are correct: `../src/images/filename.png` 3. No syntax errors in `.qmd` files

## Rendering Commands

### Render everything

```
cd website  
quarto render
```

### Render a specific file

```
cd website  
quarto render content/what-is-ai.qmd
```

### Render to specific format

```
cd website  
quarto render content/materials/presentation.qmd --to html  
quarto render content/materials/presentation.qmd --to pdf  
quarto render content/materials/presentation.qmd --to pptx
```

### Preview in browser while editing

```
cd website  
quarto preview
```

Then open: <http://localhost:4949>

## After Rendering

### Check the output

```
ls -la docs/
```

You should see: - \*.html files (standalone HTML) - \*.pdf files (PDFs) - presentation.pptx (PowerPoint) - css/style.css (copied) - images/ folder (copied) - Static HTML pages

## Test the links

1. Open `docs/index.html` in browser
2. Click “Resources” → all links should work
3. Click “Materials” → all links should work
4. Try downloading PDFs

## Deploy to GitHub Pages

```
git add docs/  
git commit -m "Render QMD files to HTML and PDF"  
git push origin main
```

GitHub Pages auto-deploys from `docs/` folder.

## Troubleshooting

### “command not found: quarto”

- Install Quarto: `brew install quarto` (or download from [quarto.org](https://quarto.org))
- Or use full path: `/usr/local/bin/quarto render`

### Images not showing in rendered HTML

- Check image paths in `.qmd` files (should be `../src/images/filename.png`)
- Verify images exist in `website/src/images/`
- Make sure `_quarto.yml` includes `resources: - src/images/`

### Links broken in rendered HTML

- Links in `website/src/resources.html` and `website/src/materials.html` should point to `../filename.html`
- These files point to the rendered output in `docs/`

### PDF generation failing

- Install a PDF rendering engine: `quarto install tinytex`
- Or use Quarto’s built-in PDF support

## PPTX presentation looking wrong

- Check that image paths use `../src/images/` format
- Use `![]` (`path/to/image.png`) syntax (not HTML `img` tags)
- Each top-level `#` heading becomes one slide

## Common Issues

### Changed a `.qmd` but it didn't update

- Quarto caches results; use `quarto render --no-cache`

### Want to rebuild everything from scratch

```
rm -rf docs/  
quarto render
```

### Different output wanted for different formats

Use format-specific YAML in the front matter:

```
format:  
  html:  
    embed-resources: true  
    toc: true  
  pdf:  
    toc: false  
    margin-left: 1in  
  pptx:  
    footer: "Your footer here"
```

## Workflow

1. **Edit `.qmd` file** → Make changes in `website/content/`
2. **Render** → `quarto render`
3. **Preview** → Open rendered HTML in browser
4. **Deploy** → `git add docs/` and push to GitHub
5. **Visit live site** → <https://michaelborck-presentations.github.io/...>

## Tips

- Use `quarto preview` while editing for live reload
- Use `quarto render --watch` to auto-render on file changes
- Keep `.qmd` files in `website/content/`, not `website/src/`
- Static HTML pages (`index.html`, `resources.html`, etc.) go in `website/src/`
- Let Quarto handle the rendering; don't edit rendered HTML files directly

## Questions?

See `WEBSITE-BUILD-GUIDE.md` for more details on the build process.